

Nonlocal convection–diffusion problems and finite element approximations[☆]

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Abstract

Nonlocal models are becoming increasingly popular in numerical simulations of important application problems, ranging from anomalous diffusion in heterogeneous environment to crack and fracture in composites. In this paper, we propose and compare some nonlocal models for convection–diffusion problems and study their finite element approximations by piecewise constant and piecewise bilinear basis functions. Our focus is on the convection-dominated case. Through analysis and numerical experiments, we demonstrate the advantage of the upwind models which are more consistent with the physical nature and allow more stable and robust numerical approximations. On the other hand, the central model may produce unphysical oscillations.

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1. Introduction

Nonlocal models given in terms of integral equations in spatial variables have received much attention in recent years [1–11], from both theoretical and computational point of view [12–23]. The modeling of central nonlocal diffusion is based on a radially symmetric kernel function, in the nonlocal operator, which describes the statistical nature of a stochastic process by assuming the probabilities of a particle moving in arbitrary directions are the same so that the processes are determined by the dependence of jump rates on jump sizes. To model more sophisticated and realistic physical process, one may need to account for convective effects (advection), which for example appear in flow conditions or dynamically changing environments [24]. Our goal is to study nonlocal-convection diffusion models (of integral-type) and their effective numerical solutions so that we can get a nonlocal extension of the diffusion–convection PDEs valid for more general jump processes and under more complex environmental conditions.

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We study the time-independent case in this work and focus on the analog of the following linear (local) convection–diffusion equation on a spatial domain Ω_s :

$$\begin{cases} -\epsilon \Delta u(\mathbf{x}) + \mathbf{b}(\mathbf{x}) \nabla u(\mathbf{x}) = f(\mathbf{x}), & \mathbf{x} \in \Omega_s, \\ u(\mathbf{x}) = g(\mathbf{x}), & \mathbf{x} \in \partial \Omega_s, \end{cases} \quad (1.1)$$

for some positive parameter $\epsilon > 0$ which is typically small in comparison with the prescribed divergence-free velocity field $\mathbf{b}(\mathbf{x})$ since we are more interested in the strong convective effects, $f(\mathbf{x})$ is a given source term and $g(\mathbf{x})$ provides boundary data. The above PDE describes the steady-state behavior of a convection–diffusion system, in which the convective flux $\mathbf{b}(\mathbf{x})u(\mathbf{x})$ is added to the diffusive flux due to Brownian motion. Problems like (1.1) enjoy many nice physical features, for example, for $f \equiv 0$ it satisfies a maximum principle, which says that the maximum of a function is to be found on the boundary of its domain. With the maximum principle, it will be easy to prove the uniqueness and stability of the solutions of related equations. Numerical schemes can also be developed to preserve such properties on the discrete level, see [25,26]. The problem (1.1) is a text-book example that has been extensively analyzed, yet its nonlocal analog has only been studied in more recent years driven by applications that range from anomalous behavior in heterogeneous environment to crack and fractures in composite materials. Studying these model equations can thus provide significant insights to more complicated processes in practice.

For nonlocal convection–diffusion models, we first recall some existing studies of nonlocal convection–diffusion models in the literature. A formulation was given in [27] that incorporated nonlinear nonlocal convection effects in a one-dimensional nonlocal balance law, but it fails to satisfy the maximum principle. It is worthy to note that there have been much studies of fractional convection–diffusion models, which may also be formulated as nonlocal models, see [28,29] and additional references cited therein. The connections between the fractional equations and nonlocal equations have been pointed out in [15] and further elaborated in [30] and [31]. One may view nonlocal equations as an alternative to fractional equations as the latter often encounter analytical difficulties when posed on a bounded domain [31]. For problems on a bounded domain, in the work [24], a formulation of nonlocal constrained value problems was developed for nonlocal convection–diffusion models, which in the time-independent case leads to

$$\begin{cases} \int_{B_\delta(\mathbf{x})} (\gamma(\mathbf{x}, \mathbf{x}')u(\mathbf{x}') - \gamma(\mathbf{x}', \mathbf{x})u(\mathbf{x})) d\mathbf{x}' = f(\mathbf{x}), & \mathbf{x} \in \Omega_s, \\ u(\mathbf{x}) = g(\mathbf{x}), & \mathbf{x} \in \Omega_c. \end{cases} \quad (1.2)$$

Here, besides the domain Ω_s which is called a solution domain, there is an additional domain Ω_c which is called constrained or interaction domain for the specification of nonlocal boundary conditions or more precisely the nonlocal solution constraints. The parameter $\delta > 0$ is the nonlocal horizon parameter, $B_\delta(\mathbf{x})$ denotes the δ -neighborhood around the point $\mathbf{x}$ which determines the range of typical nonlocal interactions, and $\gamma(\mathbf{x}, \mathbf{x}')$ denotes the kernel for the convection–diffusion term or, in probabilistic terms, a kernel characterizing the rate of jump between $\mathbf{x}$ and $\mathbf{x}'$. We refer to [24] for connections between nonlocal convection–diffusion equations and stochastic jump processes. The well-posedness of (1.2) was studied in [30]. Moreover, it is known that, under proper assumptions of the kernel function $\gamma(\mathbf{x}, \mathbf{x}')$, the nonlocal problem (1.2) converges to the following local problem:

$$\begin{cases} -\epsilon \Delta u(\mathbf{x}) + \nabla \cdot (\mathbf{b}(\mathbf{x})u(\mathbf{x})) = f(\mathbf{x}), & \mathbf{x} \in \Omega_s, \\ u(\mathbf{x}) = g(\mathbf{x}), & \mathbf{x} \in \partial \Omega_s, \end{cases} \quad (1.3)$$

as the horizon $\delta \rightarrow 0$. This is referred to as a conservative form since we have a relation

$$\int_{\Omega_s} f(\mathbf{x}) d\mathbf{x} = \int_{\partial \Omega_s} (\mathbf{b}(\mathbf{x}) \cdot \mathbf{n}(\mathbf{x})u(\mathbf{x}) - \epsilon \nabla u(\mathbf{x}) \cdot \mathbf{n}(\mathbf{x})) d\mathbf{x}$$

that describes the (local) balance law. For incompressible fields, i.e., divergence-free velocity fields $\nabla \cdot \mathbf{b} = 0$, the convection term can be reduced to $\mathbf{b}(\mathbf{x}) \cdot \nabla u(\mathbf{x})$ so that the problem (1.3) is identical to (1.1). For (1.2), maximum principle can be established for $f \equiv 0$ under suitable assumptions on the jump kernel γ , see [24].

This present work provides a couple of nonlocal versions of (1.1). In particular, one of them, which we call the upwind nonlocal convection–diffusion model, or simply the upwind model, preserves the maximum principle. The one which is named central nonlocal convection–diffusion model, or simply the central model, does not enjoy such a property. We also discuss various numerical discretizations and see in what sense the discrete maximum

principle can be preserved. It is highly relevant to the current study to recall the properties of various discretizations of associated PDE models. For linear convection dominated problems described by local PDEs, it is known that while the central difference scheme is more accurate, it is numerically unstable. On the other hand, the upwind difference scheme, which only takes the information in stream-wise direction into account, is stable and satisfies the maximum principle. As for finite element approximations, the standard Galerkin approximation is also unstable, and stabilizing terms must be added in order to increase robustness and accuracy [32]. Stabilized finite element methods for convection–diffusion problems include least-squares techniques [33,34], Galerkin least-squares techniques such as the SUPG method or the streamline diffusion method [35–37], subgrid viscosity techniques [38], and continuous interior penalty techniques [39]. In general, when developing numerical methods to solve convection-dominated problems, some special treatments are required to define the numerical flux to overcome the nonphysical oscillations. It is thus interesting and natural to examine similar questions in the nonlocal setting. Indeed, the main differences in the nonlocal models presented in this paper are reflected by whether such stream-wise information is already built into the models at the continuum level. We choose to study the upwind model and the central model side by side in order to make careful comparisons between them. For the convective term in the upwind model, we consider a one-sided kernel function so that only interactions having positive alignment with the upwind stream-wise direction is included, while in the other model, named central model, interactions in all directions are allowed. It is desirable and not hard to verify that, as $\delta \rightarrow 0$, both of these nonlocal models converge to the same local model, illustrating the fact that in a local PDE model, the natural physical notion of unwinding is no longer present which leads to one of the main modeling difficulties when encountering discontinuous velocity fields such as the case of nonlinear hyperbolic conservations laws. Suitable notions such as jump and entropy conditions have to be enforced to make sense of the physically consistent solution. In this sense, the nonlocal models provide more information on the physical reality. In addition, we note that while the studies in [24,27] are done largely for one spatial dimension, the results presented here are mostly in two spatial dimensions which also allow straightforward extensions to higher dimensional cases and dynamic models.

The rest of paper is organized as follows. In Section 2 we first present two nonlocal convection–diffusion models, namely the upwind model and the central one, and study their convergence to the conventional, local convection–diffusion equation. In Section 3 we discuss finite element approximations of the two models with piecewise constant and piecewise bilinear basis functions, and analyze the structures of their stiffness matrices in relation to the numerical stability. We also discuss the equivalence between the nonlocal conservative form and the non-conservative form for special velocity fields that may be viewed as generalizations of conventional, local divergence-free vector fields. In Section 4 we perform a series of numerical experiments to verify our theoretic results and further investigate the properties of the models. Finally, we present conclusions in Section 5.

2. Nonlocal convection–diffusion models

We first present a generic nonlocal convection–diffusion model as follows:

$$\begin{cases} -\int_{B_\delta(\mathbf{x})} \gamma_d(\mathbf{x}, \mathbf{x}')(u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' - \int_{B_\delta(\mathbf{x})} \gamma_c(\mathbf{x}, \mathbf{x}')(u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' = f(\mathbf{x}), & \mathbf{x} \in \Omega_s, \\ u(\mathbf{x}) = g(\mathbf{x}), & \mathbf{x} \in \Omega_c, \end{cases} \quad (2.1)$$

where $\gamma_d = \gamma_d(\cdot, \cdot)$ is the kernel function for the nonlocal diffusion term and $\gamma_c = \gamma_c(\cdot, \cdot)$ is the kernel function for the nonlocal convection term. The detailed forms of γ_d and γ_c are discussed next.

2.1. An upwind nonlocal convection–diffusion model

Denote the indicator function by

$$1_\Omega(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in \Omega, \\ 0, & \mathbf{x} \notin \Omega. \end{cases} \quad (2.2)$$

Let us take $D(\mathbf{z}) = 1_{B_\delta(\mathbf{0})}$ and $C(\mathbf{z}) = 1_{B_\delta(\mathbf{0})} 1_{z_1 \geq 0}$ for any $\mathbf{z} = (z_1, z_2, \dots, z_d)^T \in \mathbb{R}^d$. Then for any $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{x}' = \mathbf{x} + \mathbf{z}$, we define a nonlocal kernel function for diffusion as

$$\gamma_d(\mathbf{x}, \mathbf{x}') = \frac{\epsilon}{\delta^{d+2}} D(\mathbf{z}), \quad (2.3)$$

and the convection kernel

$$\gamma_c(\mathbf{x}, \mathbf{x}') = \gamma_c^{up}(\mathbf{x}, \mathbf{x}') = \frac{b(\mathbf{x})}{\delta^{d+1}} C(R(\mathbf{x})\mathbf{z}), \quad (2.4)$$

where $b = b(\mathbf{x}) \geq 0$ is the amplitude of the convection coefficient at $\mathbf{x}$ and $R = R(\mathbf{x})$ is a rotation matrix in $\mathbb{R}^d$ for the wind direction at $\mathbf{x}$. Note $R(\mathbf{x})$ only has $d - 1$ degrees of freedom. Obviously $\gamma_d(\mathbf{x}, \mathbf{x}')$ is a symmetric two-point function with respect to $\mathbf{x}$ and $\mathbf{x}'$, but $\gamma_c^{up}(\mathbf{x}, \mathbf{x}')$ is not.

We now specialize to the two-dimensional case for the purpose of illustration. Extensions to three and higher dimensional cases are straightforward. With regard to the assumptions of our model, the nonlocal convection kernel function is non-zero over a half-disk with radius δ (see Fig. 1), which means that only particles inside such a particular half-disk can have convective interactions with the point $\mathbf{x}$, subject to the effect of the prescribed velocity field. The angle associated with the half-disk reflects the direction of the given velocity.

Let the rotation matrix $R(\mathbf{x})$ take on the following general form

$$R(\mathbf{x}) = \begin{bmatrix} \cos \theta(\mathbf{x}) & -\sin \theta(\mathbf{x}) \\ \sin \theta(\mathbf{x}) & \cos \theta(\mathbf{x}) \end{bmatrix}, \quad (2.5)$$

where $\theta(\mathbf{x}) \in [-\pi, \pi]$ is a function of $\mathbf{x}$. Assume $u(\mathbf{x})$ is smooth, then the nonlocal diffusion term in (1.2) can be written as

$$\begin{aligned} & - \int_{B_\delta(\mathbf{x})} \gamma_d(\mathbf{x}, \mathbf{x}') (u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' \\ &= - \int_{B_\delta(\mathbf{x})} \frac{\epsilon D(\mathbf{z})}{\delta^4} (u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' \\ &= - \int_{B_\delta(\mathbf{x})} \frac{\epsilon D(\mathbf{z})}{\delta^4} \left(\frac{1}{2} (\mathbf{x}' - \mathbf{x})^T \begin{bmatrix} u_{xx} & u_{xy} \\ u_{yx} & u_{yy} \end{bmatrix} (\mathbf{x}' - \mathbf{x}) + O(\|\mathbf{x}' - \mathbf{x}\|^4) \right) d\mathbf{x}' \\ &= - \frac{\epsilon}{\delta^4} \int_0^{2\pi} \int_0^\delta \frac{1}{2} [r \cos \tau, r \sin \tau] \begin{bmatrix} u_{xx} & u_{xy} \\ u_{yx} & u_{yy} \end{bmatrix} \begin{bmatrix} r \cos \tau \\ r \sin \tau \end{bmatrix} r dr d\tau + O(\delta^2) \\ &= - \frac{\epsilon \pi}{8} \Delta u(\mathbf{x}) + O(\delta^2) \\ &\longrightarrow - \frac{\epsilon \pi}{8} \Delta u(\mathbf{x}), \quad \text{as } \delta \longrightarrow 0 \end{aligned} \quad (2.6)$$

and the nonlocal convection term in (1.2) gives us

$$\begin{aligned} & - \int_{B_\delta(\mathbf{x})} \gamma_c^{up}(\mathbf{x}, \mathbf{x}') (u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' \\ &= - \int_{B_\delta(\mathbf{x})} \frac{b(\mathbf{x})}{\delta^3} C(R(\mathbf{x})\mathbf{z}) (u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' \\ &= - \int_{B_\delta(\mathbf{x})} \frac{b(\mathbf{x})}{\delta^3} C(R(\mathbf{x})\mathbf{z}) (\nabla u(\mathbf{x}) \cdot (\mathbf{x}' - \mathbf{x}) + O(\|\mathbf{x}' - \mathbf{x}\|^2)) d\mathbf{x}' \\ &= - \frac{b(\mathbf{x})}{\delta^3} \nabla u(\mathbf{x}) \cdot \int_{B_\delta(\mathbf{x})} C(R(\mathbf{x})\mathbf{z}) (\mathbf{x}' - \mathbf{x}) d\mathbf{x}' + O(\delta) \\ &= - \frac{b(\mathbf{x})}{\delta^3} \nabla u(\mathbf{x}) \cdot \int_{-\pi/2+\theta(\mathbf{x})}^{\pi/2+\theta(\mathbf{x})} \int_0^\delta \begin{bmatrix} r \cos \tau \\ r \sin \tau \end{bmatrix} r dr d\tau + O(\delta) \\ &= - \frac{2b(\mathbf{x})}{3\delta^3} \nabla u(\mathbf{x}) \cdot \begin{bmatrix} \delta^3 \cos \theta(\mathbf{x}) \\ \delta^3 \sin \theta(\mathbf{x}) \end{bmatrix} + O(\delta) \\ &\longrightarrow \frac{2b(\mathbf{x})}{3} \begin{bmatrix} -\cos \theta(\mathbf{x}) \\ -\sin \theta(\mathbf{x}) \end{bmatrix} \cdot \nabla u(\mathbf{x}), \quad \text{as } \delta \longrightarrow 0. \end{aligned} \quad (2.7)$$

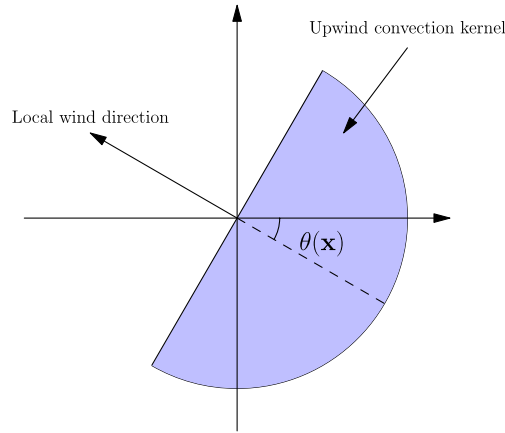


Fig. 1. Illustration of the upwind convection kernel support, $\theta(\mathbf{x}) - \frac{\pi}{2} < \tau < \theta(\mathbf{x}) + \frac{\pi}{2}$.

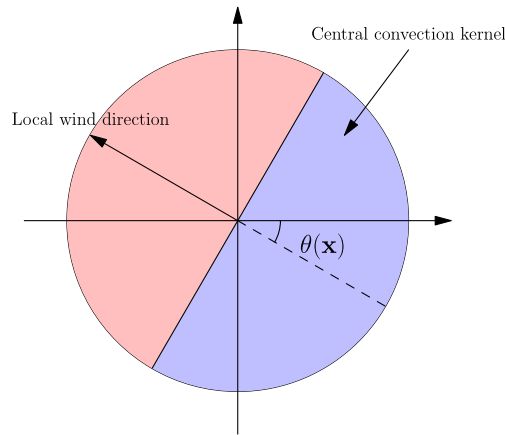


Fig. 2. Illustration of the central convection kernel support, $\theta(\mathbf{x}) - \pi < \tau < \theta(\mathbf{x}) + \pi$.

Therefore, as the horizon δ goes to 0, the nonlocal convection–diffusion model (1.2) converges formally to the following local convection–diffusion model

$$-\frac{\epsilon\pi}{8}\Delta u(\mathbf{x}) + \mathbf{b}(\mathbf{x}) \cdot \nabla u(\mathbf{x}) = f(\mathbf{x}), \quad (2.8)$$

where $\mathbf{b}(\mathbf{x}) = (v_x, v_y)^T = -\frac{2b(\mathbf{x})}{3}[\cos \theta(\mathbf{x}), \sin \theta(\mathbf{x})]^T$, i.e., $b(\mathbf{x}) = 3\sqrt{v_x^2 + v_y^2}/2$ and $\theta(\mathbf{x}) = \arctan(v_y/v_x)$. A more rigorous examination of the convergence without assuming smoothness of the solutions can be of theoretical interests as in the case of pure diffusion equations (investigated in [40] for instance), but such a study is beyond the scope of the current work.

2.2. A central nonlocal convection–diffusion model

In a similar fashion, we also can construct a nonlocal convection–diffusion model whose influence neighborhood associated with the convection is the whole disk (see Fig. 2). The diffusion term is the same as (2.3) which is defined in the nonlocal upwind convection–diffusion model, but the nonlocal kernel function for convection is different and it is given by

$$\gamma_c(\mathbf{x}, \mathbf{x}') = \gamma_c^{ct}(\mathbf{x}, \mathbf{x}') = \frac{b(\mathbf{x})}{2\delta^{d+1}}C(R(\mathbf{x})\mathbf{z}) - \frac{b(\mathbf{x})}{2\delta^{d+1}}C(-R(\mathbf{x})\mathbf{z}). \quad (2.9)$$

Then, the nonlocal convection term in (2.1) can be written as

$$\begin{aligned}
& - \int_{B_\delta(\mathbf{x})} \gamma_c^{ct}(\mathbf{x}, \mathbf{x}') (u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' \\
& = - \int_{B_\delta(\mathbf{x})} \frac{b(\mathbf{x})}{2\delta^3} (C(R(\mathbf{x})\mathbf{z}) - C(-R(\mathbf{x})\mathbf{z})) (u(\mathbf{x}') - u(\mathbf{x})) d\mathbf{x}' \\
& = - \int_{B_\delta(\mathbf{x})} \frac{b(\mathbf{x})}{2\delta^3} (C(R(\mathbf{x})\mathbf{z}) - C(-R(\mathbf{x})\mathbf{z})) \left(\nabla u(\mathbf{x}) \cdot (\mathbf{x}' - \mathbf{x}) \right. \\
& \quad \left. + \frac{1}{2}(\mathbf{x}' - \mathbf{x})^T \begin{bmatrix} u_{xx} & u_{xy} \\ u_{yx} & u_{yy} \end{bmatrix} (\mathbf{x}' - \mathbf{x}) + O(\|\mathbf{x}' - \mathbf{x}\|^3) \right) d\mathbf{x}' \\
& = - \int_{B_\delta(\mathbf{x})} \frac{b(\mathbf{x})}{2\delta^3} (C(R(\mathbf{x})\mathbf{z}) - C(-R(\mathbf{x})\mathbf{z})) (\nabla u(\mathbf{x}) \cdot (\mathbf{x}' - \mathbf{x}) + O(\|\mathbf{x}' - \mathbf{x}\|^3)) d\mathbf{x}' \\
& = - \frac{b(\mathbf{x})}{2\delta^3} \nabla u(\mathbf{x}) \cdot \int_{-\pi/2+\theta(\mathbf{x})}^{\pi/2+\theta(\mathbf{x})} \int_0^\delta \begin{bmatrix} r \cos \tau \\ r \sin \tau \end{bmatrix} r dr d\tau \\
& \quad + \frac{b(\mathbf{x})}{2\delta^3} \nabla u(\mathbf{x}) \cdot \int_{-3\pi/2+\theta(\mathbf{x})}^{-\pi/2+\theta(\mathbf{x})} \int_0^\delta \begin{bmatrix} r \cos \tau \\ r \sin \tau \end{bmatrix} r dr d\tau + O(\delta^2) \\
& = - \frac{2b(\mathbf{x})}{3\delta^3} \nabla u(\mathbf{x}) \cdot \begin{bmatrix} \delta^3 \cos \theta(\mathbf{x}) \\ \delta^3 \sin \theta(\mathbf{x}) \end{bmatrix} + O(\delta^2) \\
& = \frac{2b(\mathbf{x})}{3} \begin{bmatrix} -\cos \theta(\mathbf{x}) \\ -\sin \theta(\mathbf{x}) \end{bmatrix} \cdot \nabla u(\mathbf{x}), \quad \text{as } \delta \rightarrow 0.
\end{aligned} \tag{2.10}$$

Hence, as $\delta \rightarrow 0$, the nonlocal problem (1.2) reduces to the same local convection–diffusion problem as shown in (2.8).

3. Stiffness matrices and stability analysis of finite element approximations

Without loss of generality, to do the stability study, we assume $b(\mathbf{x}) \equiv B$ and $\theta(\mathbf{x}) \equiv \theta$ where B and θ are constants. We perform analysis of the standard finite element approximations of both models on uniform meshes as illustration. Our results extend similar calculations done in [17] for a one-dimensional nonlocal pure diffusion models. In this paper, the calculations are done in two dimensions.

3.1. The upwind nonlocal convection–diffusion model

Assume the finite element space is spanned by the basis $\mathcal{T} = \{\phi_1, \phi_2, \dots, \phi_N\}$ associated with the nodes $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ over a uniform Cartesian lattice. Then according to (2.1), the (i, j) -th element of the resulting stiffness matrix M for the upwind nonlocal model is determined by

$$M_{i,j} = -\frac{\epsilon}{\delta^4} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x})} (\phi_j(\mathbf{x}') - \phi_j(\mathbf{x})) d\mathbf{x}' d\mathbf{x} - \frac{B}{\delta^3} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x})} (\phi_j(\mathbf{x}') - \phi_j(\mathbf{x})) d\mathbf{x}' d\mathbf{x}, \tag{3.1}$$

where S_i denotes the support for the basis function ϕ_i and $B_\delta^\theta(\mathbf{x})$ denotes the shaded half-disk region as shown in Fig. 1.

Let us split the above matrix M into two parts, the diffusion part M^D and the convection part M^C , then it holds that

$$\begin{aligned}
M_{i,j}^D & = -\frac{\epsilon}{\delta^4} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x})} (\phi_j(\mathbf{x}') - \phi_j(\mathbf{x})) d\mathbf{x}' d\mathbf{x} \\
& = -\frac{\epsilon}{\delta^4} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} + \frac{\epsilon \|B_\delta\|}{\delta^4} \int_{S_i \cap S_j} \phi_i(\mathbf{x}) \phi_j(\mathbf{x}) d\mathbf{x},
\end{aligned} \tag{3.2}$$

$$\begin{aligned}
M_{i,j}^C &= -\frac{B}{\delta^3} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x})} (\phi_j(\mathbf{x}') - \phi_j(\mathbf{x})) d\mathbf{x}' d\mathbf{x} \\
&= -\frac{B}{\delta^3} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x})} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} + \frac{B}{\delta^3} \int_{S_i} \phi_i(\mathbf{x}) \phi_j(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x})} d\mathbf{x}' d\mathbf{x}, \\
&= -\frac{B}{\delta^3} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} + \frac{B \|B_\delta\|}{2\delta^3} \int_{S_i \cap S_j} \phi_i(\mathbf{x}) \phi_j(\mathbf{x}) d\mathbf{x},
\end{aligned} \tag{3.3}$$

where $\|\cdot\|$ represents the measure of area. It is easy to verify that, for any i ,

$$\begin{aligned}
\sum_j M_{i,j}^D &= -\frac{\epsilon}{\delta^4} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x})} d\mathbf{x}' d\mathbf{x} + \frac{\epsilon \|B_\delta\|}{\delta^4} \int_{S_i} \phi_i(\mathbf{x}) \left(\sum_j \phi_j(\mathbf{x}) \right) d\mathbf{x} \\
&= -\frac{\epsilon}{\delta^4} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x})} \sum_j \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} + \frac{\epsilon \|B_\delta\|}{\delta^4} \int_{S_i} \phi_i(\mathbf{x}) d\mathbf{x} \\
&= 0.
\end{aligned}$$

Similarly, $\sum_j M_{i,j}^C = 0$ also holds. Thus, we get

$$\sum_j M_{i,j} = \sum_j (M_{i,j}^D + M_{i,j}^C) = 0. \tag{3.4}$$

For any diagonal entry M_{ii} , by (3.2) and (3.3), we first see that

$$\begin{aligned}
M_{i,i}^D &= -\frac{\epsilon}{\delta^4} \int_{\Omega} \int_{\Omega} \phi_i(\mathbf{x}) (\phi_i(\mathbf{x}') - \phi_i(\mathbf{x})) 1_{B_\delta(\mathbf{x})} d\mathbf{x}' d\mathbf{x} \\
&= -\frac{\epsilon}{\delta^4} \int_{\Omega} \int_{\Omega} \phi_i(\mathbf{x}') (\phi_i(\mathbf{x}) - \phi_i(\mathbf{x}')) 1_{B_\delta(\mathbf{x})} d\mathbf{x}' d\mathbf{x}
\end{aligned}$$

where a variable interchange is used to get the second equation. Thus we have,

$$-\frac{\epsilon}{\delta^4} \left(\int_{\Omega} \int_{\Omega} (\phi_i(\mathbf{x}) + \phi_i(\mathbf{x}')) (\phi_i(\mathbf{x}) - \phi_i(\mathbf{x}')) 1_{B_\delta(\mathbf{x})} d\mathbf{x}' d\mathbf{x} \right) = 0.$$

That is,

$$\begin{aligned}
M_{i,i}^D &= -\frac{\epsilon}{\delta^4} \int_{\Omega} \int_{\Omega} \frac{2\phi_i(\mathbf{x}) - (\phi_i(\mathbf{x}) + \phi_i(\mathbf{x}'))}{2} (\phi_i(\mathbf{x}') - \phi_i(\mathbf{x})) 1_{B_\delta(\mathbf{x})} d\mathbf{x}' d\mathbf{x} \\
&= -\frac{\epsilon}{2\delta^4} \int_{\Omega} \int_{\Omega} (\phi_i(\mathbf{x}) - \phi_i(\mathbf{x}')) (\phi_i(\mathbf{x}') - \phi_i(\mathbf{x})) 1_{B_\delta(\mathbf{x})} d\mathbf{x}' d\mathbf{x} > 0.
\end{aligned}$$

Similarly, we consider convective parts.

$$\begin{aligned}
M_{i,i}^C &= -\frac{B}{\delta^3} \int_{\Omega} \int_{\Omega} (\phi_i(\mathbf{x}) \phi_i(\mathbf{x}') - \phi_i(\mathbf{x}) \phi_i(\mathbf{x})) 1_{B_\delta^\theta(\mathbf{x})} d\mathbf{x}' d\mathbf{x} \\
&= -\frac{B}{\delta^3} \int_{\Omega} \int_{\Omega} (\phi_i(\mathbf{x}) \phi_i(\mathbf{x}') - \phi_i(\mathbf{x}') \phi_i(\mathbf{x}')) 1_{\tilde{B}_\delta^\theta(\mathbf{x})} d\mathbf{x}' d\mathbf{x}
\end{aligned}$$

where $\tilde{B}_\delta^\theta(\mathbf{x})$ denotes the corresponding half-disk region symmetric to $B_\delta^\theta(\mathbf{x})$ in Fig. 1. Adding the two expressions in the above equation together, we have

$$\begin{aligned}
2M_{i,i}^C &= -\frac{B}{\delta^3} \int_{\Omega} \int_{\Omega} (\phi_i(\mathbf{x}) \phi_i(\mathbf{x}') 1_{B_\delta(\mathbf{x})} - \phi_i(\mathbf{x}) \phi_i(\mathbf{x}) 1_{B_\delta^\theta(\mathbf{x})} - \phi_i(\mathbf{x}') \phi_i(\mathbf{x}') 1_{\tilde{B}_\delta^\theta(\mathbf{x})}) d\mathbf{x}' d\mathbf{x} \\
&= -\frac{B}{\delta^3} \int_{\Omega} \int_{\Omega} (\phi_i(\mathbf{x}) \phi_i(\mathbf{x}') 1_{B_\delta(\mathbf{x})} - \phi_i(\mathbf{x}) \phi_i(\mathbf{x}) 1_{B_\delta(\mathbf{x})}) d\mathbf{x}' d\mathbf{x} \\
&= \frac{B\delta}{\epsilon} M_{i,i}^D > 0.
\end{aligned}$$

Thus we have that the main diagonal entry satisfies $M_{i,i} = (1 + \frac{B\delta}{2\epsilon}) M_{i,i}^D > 0$.

We next focus on the off-diagonal entries.

3.1.1. Discretization with piecewise constant basis functions

First we consider the finite element discretization using piecewise constant basis functions. For any $i \neq j$, since S_i and S_j do not overlap in the constant element discretization, we easily get the off-diagonal entries

$$M_{i,j} = -\frac{\epsilon}{\delta^4} \left(\int_{S_i} \int_{B_\delta(\mathbf{x}) \cap S_j} d\mathbf{x}' d\mathbf{x} \right) - \frac{B}{\delta^3} \left(\int_{S_i} \int_{B_\delta^\theta(\mathbf{x}) \cap S_j} d\mathbf{x}' d\mathbf{x} \right) \leq 0, \quad (3.5)$$

thus we obtain the following result:

Theorem 1. *For the finite element approximation of the upwind nonlocal convection–diffusion model using piecewise constant basis functions, the resulting stiffness matrix M is an M -matrix that guarantees that the approximation is numerically stable and satisfies the discrete maximum principle.*

We note that an M -matrix is a matrix with nondiagonal entries being nonpositive and eigenvalues whose real parts are positive. The M -matrix property leads to the discrete maximum principle that provides the L_∞ stability, we refer to [17] for detailed discussions on these notions for the special one-dimensional case without convection. Indeed, the above theorem remains valid for $B = 0$, that is, the pure-diffusion case.

3.1.2. Discretization with piecewise bilinear basis functions

To consider $M_{i,j}$ with $i \neq j$, we first note that for those ϕ_j such that S_j and S_i do not overlap, similar to the case of piecewise constant basis function, we have $M_{i,j} \leq 0$. That is, $M_{i,j}$ is always nonpositive if the node $\mathbf{x}_j$ is not an immediate neighbor of $\mathbf{x}_i$.

For those ϕ_j that S_j and S_i do overlap, we divide them into two types: the first are those S_j which overlap exactly half of S_i and denote such ϕ_j by ϕ_{jh} , and the other are those S_j which only overlap one quarter of S_i and denote such ϕ_j by ϕ_{jq} . Then we have

$$\begin{aligned} M_{i,jh} &= -\frac{\epsilon}{\delta^4} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x}) \cap S_{jh}} \phi_{jh}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \|B_\delta\| \int_{S_i \cap S_{jh}} \phi_i(\mathbf{x}) \phi_{jh}(\mathbf{x}) d\mathbf{x} \right) \\ &\quad - \frac{B}{\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_{jh}} \phi_{jh}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \frac{\|B_\delta\|}{2} \int_{S_i \cap S_{jh}} \phi_i(\mathbf{x}) \phi_{jh}(\mathbf{x}) d\mathbf{x} \right) \\ &= -\frac{\epsilon h^2}{\delta^4} \left(\tilde{b} h^2 - \frac{\pi \delta^2}{9} \right) - \frac{B h^2}{\delta^3} \left(\tilde{c} h^2 - \frac{\pi \delta^2}{18} \right) \\ &= -\left(\tilde{b} \epsilon + \tilde{c} B \delta - \frac{2\epsilon \pi + B \delta \pi}{18} \frac{\delta^2}{h^2} \right) \frac{h^4}{\delta^4}, \end{aligned} \quad (3.6)$$

where

$$\tilde{b} = \frac{\int_{B_\delta(\mathbf{x}) \cap S_{jh}} \phi_{jh}(\mathbf{x}') d\mathbf{x}'}{\int_{S_{jh}} \phi_{jh}(\mathbf{x}') d\mathbf{x}'}, \quad \tilde{c} = \frac{\int_{B_\delta^\theta(\mathbf{x}) \cap S_{jh}} \phi_{jh}(\mathbf{x}') d\mathbf{x}'}{\int_{S_{jh}} \phi_{jh}(\mathbf{x}') d\mathbf{x}'}.$$

To guarantee that $M_{i,j}$ is an M -matrix, we need to impose some restriction on the measure of the neighborhood. In order to guarantee that $M_{i,jh}$ is non-positive, let $\delta/h = K$, it must hold

$$\tilde{b} \epsilon + \tilde{c} B \delta - \frac{2\epsilon \pi + B \delta \pi}{18} K^2 \geq 0 \implies K \leq \sqrt{\frac{\tilde{b} \epsilon + \tilde{c} B \delta}{2\epsilon + B \delta} \frac{18}{\pi}},$$

and thus a sufficient condition is that

$$K \leq \sqrt{\frac{\tilde{b} \epsilon}{2\epsilon + B \delta} \frac{18}{\pi}}, \quad (3.7)$$

since c could be zero. At the same time,

$$\begin{aligned}
 M_{i,jq} &= -\frac{\epsilon}{\delta^4} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x}) \cap S_{jq}} \phi_{jq}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \|B_\delta\| \int_{S_i \cap S_{jq}} \phi_i(\mathbf{x}) \phi_{jq}(\mathbf{x}) d\mathbf{x} \right) \\
 &\quad - \frac{B}{\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_{jq}} \phi_{jq}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \frac{\|B_\delta\|}{2} \int_{S_i \cap S_{jq}} \phi_i(\mathbf{x}) \phi_{jq}(\mathbf{x}) d\mathbf{x} \right) \\
 &= -\frac{\epsilon h^2}{\delta^4} \left(\hat{b} h^2 - \frac{\pi \delta^2}{36} \right) - \frac{B h^2}{\delta^3} \left(\hat{c} h^2 - \frac{\pi \delta^2}{72} \right) \\
 &= -\left(\hat{b}\epsilon + \hat{c}B\delta - \frac{2\epsilon\pi + B\delta\pi}{72} \frac{\delta^2}{h^2} \right) \frac{h^4}{\delta^4},
 \end{aligned} \tag{3.8}$$

where

$$\hat{b} = \frac{\int_{B_\delta(\mathbf{x}) \cap S_{jq}} \phi_{jq}(\mathbf{x}') d\mathbf{x}'}{\int_{S_{jq}} \phi_{jq}(\mathbf{x}') d\mathbf{x}'}, \quad \hat{c} = \frac{\int_{B_\delta^\theta(\mathbf{x}) \cap S_{jq}} \phi_{jq}(\mathbf{x}') d\mathbf{x}'}{\int_{S_{jq}} \phi_{jq}(\mathbf{x}') d\mathbf{x}'}.$$

To make $M_{i,jq} \leq 0$ it is sufficient to require

$$K \leq \sqrt{\frac{\hat{b}\epsilon}{2\epsilon + \delta B} \frac{72}{\pi}}. \tag{3.9}$$

Combining (3.7) and (3.9), we require that

$$K \leq \sqrt{\frac{b_0\epsilon}{2\epsilon + \delta B} \frac{18}{\pi}}, \tag{3.10}$$

where $b_0 = \min\{\tilde{b}, 4\hat{b}\}$.

Theorem 2. *For the finite element approximation of the upwind nonlocal convection–diffusion model using piecewise bilinear basis functions, we have that*

- (i) *if the node $\mathbf{x}_j$ is not an immediate neighbor of $\mathbf{x}_i$ for $i \neq j$, then the entry $M_{i,j}$ in the stiffness matrix M is always non-positive;*
- (ii) *if $K \leq \sqrt{\frac{b_0\epsilon}{2\epsilon + \delta B} \frac{18}{\pi}}$, then the resulting stiffness matrix M is an M -matrix.*

We first note that the bound on K is satisfied in the local limit (that is $\delta \rightarrow 0$) so the M -matrix property in the local limit holds for both piecewise constant and piecewise bilinear basis functions. The result generalizes the special case for pure-diffusion problems considered in [17]. We also note that all the constraints in Theorem 2 are only sufficient conditions for numerical stability as discussed in [17] since one may have stabilities in other norms even if the discrete maximum principle does not hold.

3.2. The central nonlocal convection–diffusion model

For the central nonlocal model, the entries of the resulting stiffness matrix M are determined by

$$\begin{aligned}
 M_{i,j} &= -\frac{\epsilon}{\delta^4} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x})} (\phi_j(\mathbf{x}') - \phi_j(\mathbf{x})) d\mathbf{x}' d\mathbf{x} \\
 &\quad - \frac{B}{2\delta^3} \int_{S_i} \phi_i(\mathbf{x}) \left(\int_{B_\delta^\theta(\mathbf{x})} (\phi_j(\mathbf{x}') - \phi_j(\mathbf{x})) d\mathbf{x}' - \int_{\tilde{B}_\delta^\theta(\mathbf{x})} (\phi_j(\mathbf{x}') - \phi_j(\mathbf{x})) d\mathbf{x}' \right) d\mathbf{x},
 \end{aligned}$$

where $B_\delta^\theta(\mathbf{x})$ denotes the dark blue shaded region as in Fig. 2, and $\tilde{B}_\delta^\theta(\mathbf{x})$ denotes the light red shaded region.

The diffusion part $M_{i,j}^D$ is the same as in the upwind nonlocal model, but for the convection part $M_{i,j}^C$, we have

$$\begin{aligned}
 M_{i,j}^C &= -\frac{B}{2\delta^3} \int_{S_i} \phi_i(\mathbf{x}) \left(\int_{B_\delta^\theta(\mathbf{x})} (\phi_j(\mathbf{x}') - \phi_j(\mathbf{x})) d\mathbf{x}' - \int_{\tilde{B}_\delta^\theta(\mathbf{x})} (\phi_j(\mathbf{x}') - \phi_j(\mathbf{x})) d\mathbf{x}' \right) d\mathbf{x} \\
 &= -\frac{B}{2\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \frac{\|B_\delta\|}{2} \int_{S_i \cap S_j} \phi_i(\mathbf{x}) \phi_j(\mathbf{x}) d\mathbf{x} \right. \\
 &\quad \left. - \int_{S_i} \phi_i(\mathbf{x}) \int_{\tilde{B}_\delta^\theta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} + \frac{\|\tilde{B}_\delta\|}{2} \int_{S_i \cap S_j} \phi_i(\mathbf{x}) \phi_j(\mathbf{x}) d\mathbf{x} \right) \\
 &= -\frac{B}{2\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \int_{S_i} \phi_i(\mathbf{x}) \int_{\tilde{B}_\delta^\theta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right) \\
 &= -\frac{B}{2\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} + \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right. \\
 &\quad \left. - \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \int_{S_i} \phi_i(\mathbf{x}) \int_{\tilde{B}_\delta^\theta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right) \\
 &= -\frac{B}{\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \frac{1}{2} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x}) \cap S_j} \phi_j(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right). \tag{3.11}
 \end{aligned}$$

For the diagonal entry, the diffusion part M^D is the same as in the nonlocal upwind model. We investigate the convection part M^C . For any δ , it holds that

$$\begin{aligned}
 M_{i,i}^C &= -\frac{B}{\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_i} \phi_i(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right) + \frac{B}{2\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x}) \cap S_i} \phi_i(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right) \\
 &= 0, \tag{3.12}
 \end{aligned}$$

thus

$$M_{i,i} = M_{i,i}^D > 0.$$

We discuss the off-diagonal entries separately.

3.2.1. Discretization with piecewise constant basis functions

For any $i \neq j$, we can easily get

$$M_{i,j} = -\left(\frac{\epsilon}{\delta^4} - \frac{B}{2\delta^3}\right) \int_{S_i} \int_{B_\delta(\mathbf{x}) \cap S_j} d\mathbf{x}' d\mathbf{x} - \frac{B}{\delta^3} \int_{S_i} \int_{B_\delta^\theta(\mathbf{x}) \cap S_j} d\mathbf{x}' d\mathbf{x}. \tag{3.13}$$

To make $M_{i,j} \leq 0$ for $i \neq j$, it is required that

$$\frac{\epsilon}{\delta^4} - \frac{B}{2\delta^3} \geq 0 \implies \frac{\delta B}{\epsilon} \leq 2. \tag{3.14}$$

We remark that this is significantly different than the situation (3.5) of the upwind nonlocal model where no condition on the coefficients is needed. We summarize the results below:

Theorem 3. *For the finite element approximation of the central nonlocal convection–diffusion model using piecewise constant basis functions, if $\delta B \leq 2\epsilon$, then the resulting stiffness matrix M is an M -matrix.*

3.2.2. Discretization with piecewise bilinear basis functions

Now we consider the case of piecewise bilinear basis function. Similar as the upwind cases, we first consider those ϕ_j such that S_j and S_i do not overlap. We denote them by ϕ_{jn} . Then

$$M_{i,jn} = -\left(\frac{\epsilon}{\delta^4} - \frac{B}{2\delta^3}\right) \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x}) \cap S_{jn}} \phi_{jn}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \frac{B}{\delta^3} \int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_{jn}} \phi_{jn}(\mathbf{x}) d\mathbf{x}' d\mathbf{x}. \quad (3.15)$$

The condition (3.14) is also needed to obtain $M_{i,jn} \leq 0$. We remark that this is again significantly different from the upwind nonlocal model where $M_{i,jn}$ is always non-positive.

As for the other off-diagonal entries, we have

$$\begin{aligned} M_{i,jh} &= -\frac{\epsilon}{\delta^4} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x}) \cap S_{jh}} \phi_{jh}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \|B_\delta\| \int_{S_i \cap S_{jh}} \phi_i(\mathbf{x}) \phi_{jh}(\mathbf{x}) d\mathbf{x} \right) \\ &\quad - \frac{B}{\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_{jh}} \phi_{jh}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right) + \frac{B}{2\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x}) \cap S_{jh}} \phi_{jh}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right) \\ &= -\frac{\epsilon h^2}{\delta^4} \left(\tilde{b} h^2 - \frac{\pi \delta^2}{9} \right) - \frac{B h^4}{\delta^3} \left(\tilde{c} - \frac{\tilde{b}}{2} \right) \\ &= -\left(\tilde{b} \epsilon + \tilde{c} \delta B - \frac{\tilde{b} \delta B}{2} - \frac{\epsilon \pi \delta^2}{9 h^2} \right) \frac{h^4}{\delta^4}. \end{aligned} \quad (3.16)$$

To guarantee $M_{i,jh} \leq 0$, we get

$$\tilde{b} \epsilon + \tilde{c} \delta B - \frac{\tilde{b} \delta B}{2} - \frac{\epsilon \pi}{9} K^2 \geq 0 \implies K \leq \sqrt{\frac{2\tilde{b} \epsilon + 2\tilde{c} \delta B - \tilde{b} \delta B}{2\epsilon} \frac{9}{\pi}},$$

under the condition (3.14), and thus

$$K \leq \sqrt{\frac{2\tilde{b} \epsilon - \tilde{b} \delta B}{2\epsilon} \frac{9}{\pi}}. \quad (3.17)$$

We also have

$$\begin{aligned} M_{i,jq} &= -\frac{\epsilon}{\delta^4} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x}) \cap S_{jq}} \phi_{jq}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} - \|B_\delta\| \int_{S_i \cap S_{jq}} \phi_i(\mathbf{x}) \phi_{jq}(\mathbf{x}) d\mathbf{x} \right) \\ &\quad - \frac{B}{\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta^\theta(\mathbf{x}) \cap S_{jq}} \phi_{jq}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right) + \frac{B}{2\delta^3} \left(\int_{S_i} \phi_i(\mathbf{x}) \int_{B_\delta(\mathbf{x}) \cap S_{jq}} \phi_{jq}(\mathbf{x}') d\mathbf{x}' d\mathbf{x} \right) \\ &= -\frac{\epsilon h^2}{\delta^4} \left(\hat{b} h^2 - \frac{\pi \delta^2}{36} \right) - \frac{B h^4}{\delta^3} \left(\hat{c} - \frac{\hat{b}}{2} \right) \\ &= -\left(\hat{b} \epsilon + \hat{c} \delta B - \frac{\hat{b} \delta B}{2} - \frac{\epsilon \pi}{36} K^2 \right) \frac{h^4}{\delta^4}. \end{aligned} \quad (3.18)$$

Similarly, to make $M_{i,jq} \leq 0$ we need that

$$K \leq \sqrt{\frac{2\hat{b} \epsilon - \hat{b} \delta B}{2\epsilon} \frac{36}{\pi}}. \quad (3.19)$$

Combining (3.17) and (3.19), we have that

$$K \leq \sqrt{\frac{b_0(2\epsilon - \delta B)}{2\epsilon}} \frac{9}{\pi}, \quad (3.20)$$

where $b_0 = \min\{\tilde{b}, 4\hat{b}\}$.

Thus we conclude that:

Theorem 4. *For the finite element approximation of the central nonlocal convection–diffusion model using piecewise bilinear basis functions, we have that*

- (i) *if the node $\mathbf{x}_j$ is not an immediate neighbor of $\mathbf{x}_i$ for $i \neq j$ and $\delta B \leq 2\epsilon$, then the entry $M_{i,j}$ in the stiffness matrix M is non-positive;*
- (ii) *if $\delta B \leq 2\epsilon$ and $K \leq \sqrt{\frac{b_0(2\epsilon - \delta B)}{2\epsilon}} \frac{9}{\pi}$, then the resulting stiffness matrix M is an M -matrix.*

In contrast to the upwind case, we see that the results for the central nonlocal model are valid only where there is a strong diffusion effect with a relatively small B . Indeed, for $B = 0$, both Theorems 3 and 4 are consistent with the known calculations made in [17] for one dimensional nonlocal pure-diffusion models.

3.3. The conservative form

We know that (2.1) is a nonlocal analog of the local convection–diffusion in non-conservative form. Since it is equivalent to the conservation form for divergence-free velocity, we like to present a conservative form as a nonlocal analog for the so-called nonlocal divergence free field. First, a conservative form can be easily written as

$$-\int_{B_\delta(\mathbf{x})} (\gamma_d(\mathbf{x}, \mathbf{x}')u(\mathbf{x}') - \gamma_d(\mathbf{x}', \mathbf{x})u(\mathbf{x})) d\mathbf{x}' - \int_{B_\delta(\mathbf{x})} (\gamma_c(\mathbf{x}, \mathbf{x}')u(\mathbf{x}') - \gamma_c(\mathbf{x}', \mathbf{x})u(\mathbf{x})) d\mathbf{x}' = f(\mathbf{x}). \quad (3.21)$$

By the symmetry of γ_d , to see the equivalence of (2.1) and (3.21), the condition needed is

$$\int_{B_\delta(\mathbf{x})} (\gamma_c(\mathbf{x}', \mathbf{x}) - \gamma_c(\mathbf{x}, \mathbf{x}')) d\mathbf{x}' = 0.$$

Since γ_c is defined by $b(\mathbf{x})$ and $\theta(\mathbf{x})$, this provides a condition for the equivalence. Indeed, it can be shown that as $\delta \rightarrow 0$, the above equation, in the limit, reduces to $\nabla \cdot \mathbf{b}(\mathbf{x}) = 0$. Thus, we may indeed view the above condition as a nonlocal analog of the divergence-free condition.

4. Numerical experiments

In this section both finite element approximations with piecewise constant and piecewise bilinear basis functions are applied to solve the nonlocal convection–diffusion problems (2.1). We compare the performance of these two models and their finite element approximations to further verify our analyses given in Section 3. Two types of convergence behaviors, K -convergence and δ -convergence [41], will be investigated: for K -convergence, we fix the length of horizon and keep refining the mesh (thus enlarging K), and for δ -convergence, we instead fix the ratio between the horizon $\delta = Kh$ and h during the mesh refinement (thus diminishing δ). While the former gives standard numerical convergence tests, the latter is of particular interests in validating the local limit since it tests, by setting $\delta \rightarrow 0$, if the results of nonlocal models are consistent with that produced directly via local models.

4.1. Tests for K -convergence

For the K -convergence, we take the view that the horizon δ is a material parameter so that its value is independent of the mesh size. While keeping δ fixed, with $h \rightarrow 0$, the approximate numerical solutions are expected to converge to the exact analytical solution of the nonlocal model.

Example 1. In this example let us take $\Omega_s = [0, 1] \times [0, 1]$, the exact solution $u(x, y) = x^4 y^2$ and set the horizon to be fixed as $\delta = 0.2$. The convection kernel is given with direction $\theta(x, y) = \frac{\pi}{18}$ and amplitude $b(x, y) = 1$. We will

Table 1

Numerical results on errors and convergence rates of finite element approximations of the upwind nonlocal model in Example 1.

$\delta = 0.2$								
h	$\epsilon = 1.0$				$\epsilon = 10^{-5}$			
	p.w. constant		p.w. bilinear		p.w. constant		p.w. bilinear	
	$\ e^h\ _{L^2}$	CR	$\ e^h\ _{L^2}$	CR	$\ e^h\ _{L^2}$	CR	$\ e^h\ _{L^2}$	CR
1/10	4.5737e-02	–	5.6863e-03	–	2.2814e-02	–	8.0574e-03	–
1/20	2.2343e-02	1.03	1.8614e-03	1.61	1.1227e-02	1.02	2.1315e-03	1.92
1/40	1.0721e-02	1.06	4.8481e-04	1.94	5.6528e-03	0.99	6.6779e-04	1.67
1/80	5.2640e-03	1.03	1.2712e-04	1.93	2.8989e-03	0.96	1.7969e-04	1.89
1/160	2.6131e-03	1.01	3.4712e-05	1.87	1.4392e-03	1.01	4.9391e-05	1.86

Table 2

Numerical results on errors and convergence rates of finite element approximations of the central nonlocal model in Example 1.

$\delta = 0.2$				
h	$\epsilon = 1.0$			
	p.w. constant		p.w. bilinear	
	$\ e^h\ _{L^2}$	CR	$\ e^h\ _{L^2}$	CR
1/10	5.3388e-02	–	6.3999e-03	–
1/20	2.4268e-02	1.13	1.8614e-03	1.78
1/40	1.1616e-02	1.06	4.9985e-04	1.90
1/80	5.2640e-03	1.14	1.2863e-04	1.96
1/160	2.6012e-03	1.02	3.4125e-05	1.91

test the case with diffusion coefficient $\epsilon = 1$ and $\epsilon = 10^{-5}$. The right-hand side $f(x, y)$ is found by the method of manufactured solutions, and $g(x, y) = u(x, y)$ in Ω_c .

We solve problem (2.1) with the two nonlocal models: the upwind model and the central model over the $N \times N$ ($h = 1/N$) square meshes with $N = 10, 20, 40, 80, 160$. Table 1 lists the numerical results of the finite element approximations of the upwind nonlocal convection–diffusion model, that demonstrate the stability and convergence of finite element approximations of the model for both $\epsilon = 1$ and $\epsilon = 10^{-5}$ (convection-dominated) cases. It is also easy to see that the convergence rates are optimal, i.e., first-order for the piecewise constant basis function discretization and second-order for the piecewise bilinear basis function discretization. This verifies our analysis summarized in Theorem 1 for the finite element approximation with piecewise constant basis functions. It is also worth noting that although the constraints in Theorem 2 are not satisfied for the convection-dominated case, the finite element approximation with piecewise bilinear basis functions seems still very stable and performs well (see the left figure of Fig. 3 for an illustration of distribution of L_2 errors).

In Table 2, we present the results for the central model with $\epsilon = 1$ and the convergence rates are also optimal in this case for both piecewise constant and bilinear basis function discretizations. However, for the case of $\epsilon = 10^{-5}$ (convection-dominated), both the piecewise constant and piecewise bilinear approximations fail to converge under the given meshes, in fact, they generate very strong unphysical oscillations as observed in the right figure of Fig. 3 since the condition required in Theorem 4 is violated.

Example 2. In the second example, the nonlocal convection–diffusion model with variable wind will be tested. We still take $\Omega_s = [0, 1] \times [0, 1]$, the exact solution $u(x, y) = x^4 y^2$ and the horizon $\delta = 0.2$. We here set the magnitude $b(x, y) = \sqrt{(1+x)^2 + (1-y)^2}$ and the direction $\theta = \arctan(\frac{1-y}{1+x})$ (it is easy to verify that $\nabla \cdot \mathbf{b}(x, y) = 0$ for the corresponding local equation). The right-hand $f(x, y)$ and the boundary condition $g(x, y)$ are imposed correspondingly.

We again perform a series of numerical experiments to observe the K -convergence of the finite element approximations of the two nonlocal convection–diffusion models with the variable velocity field. Numerical results for different cases are reported in Tables 3 and 4 for two different nonlocal models. We observe that all the stability and convergence behaviors are similar to those in Example 1.

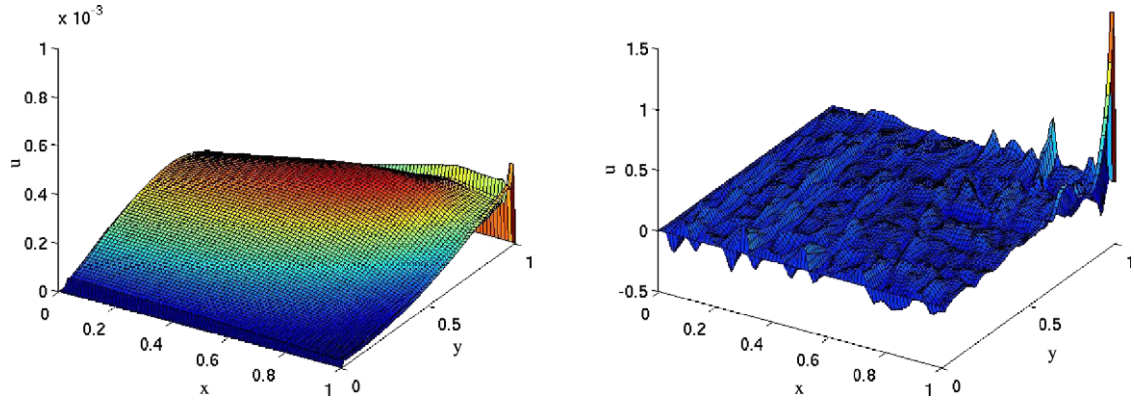


Fig. 3. Distribution of L_2 errors of finite element approximations of the upwind model (left) and the central model (right) with $\epsilon = 10^{-5}$ using piecewise bilinear basis functions on a 80×80 mesh in Example 1.

Table 3

Numerical results on errors and convergence rates of finite element approximations of the upwind nonlocal model in Example 2.

$\delta = 0.2$								
h	$\epsilon = 1.0$				$\epsilon = 10^{-5}$			
	p.w. constant		p.w. bilinear		p.w. constant		p.w. bilinear	
	$\ e^h\ _{L^2}$	CR	$\ e^h\ _{L^2}$	CR	$\ e^h\ _{L^2}$	CR	$\ e^h\ _{L^2}$	CR
1/10	4.8739e-02	–	5.2777e-03	–	2.3762e-02	–	7.6865e-03	–
1/20	2.4131e-02	1.01	1.5424e-04	1.77	1.1439e-02	1.00	2.1210e-03	1.86
1/40	1.2091e-02	0.99	4.7266e-04	1.71	5.6612e-03	1.00	6.2541e-04	1.76
1/80	6.0096e-03	1.01	1.3096e-04	1.86	2.8236e-03	0.99	1.8076e-04	1.79

Table 4

Numerical results on errors and convergence rates of finite element approximations of the central nonlocal model in Example 2.

$\delta = 0.2$				
N	$\epsilon = 1.0$			
	p.w. constant		p.w. bilinear	
	$\ e^h\ _{L^2}$	CR	$\ e^h\ _{L^2}$	CR
1/10	2.4410e-02	–	3.7063e-03	–
1/20	1.1468e-02	1.08	1.0424e-03	1.83
1/40	5.6124e-03	1.03	3.0147e-04	1.79
1/80	2.8011e-03	1.00	8.6123e-05	1.81

Example 3. In the third example, we again pick $\Omega_s = [0, 1] \times [0, 1]$, $B = 1$, $\theta = \frac{\pi}{18}$, but impose a Dirichlet volume-constraint condition as shown in Fig. 4: in the blue region, set $u(x, y) = 1$ and in the white region, set $u(x, y) = 0$. We choose the right-hand side $f(x, y) = 0$, and test both cases with the diffusion coefficient $\epsilon = 1$ and $\epsilon = 10^{-5}$. Note that the exact solutions are unknown in this example. Our focus is to examine the numerical stability issues of the discrete approximations.

We solve the nonlocal models by the finite element approximations on a 80×80 mesh. Fig. 5 shows the approximate solutions for the case of $\epsilon = 1$. Due to the strong diffusion, it is easy to see that both models solved by either piecewise constant or piecewise bilinear element discretizations give very similar and stable results for which the maximum principle seems to remain satisfied.

In Fig. 6, we show the simulation results for the convection-dominated case ($\epsilon = 10^{-5}$). According to the nonlocal boundary condition (volumetric constraint) we impose, the exact solutions should have a boundary layer effect close to $x = 0$ and $y = 0$, i.e., a rapid change from 1 to 0. As we see from Fig. 6, the upwind nonlocal model works well with finite element discretizations with either piecewise constant or piecewise bilinear basis functions, though

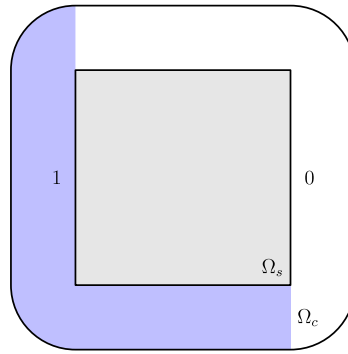


Fig. 4. The domain and volume-constraint conditions for Example 3.

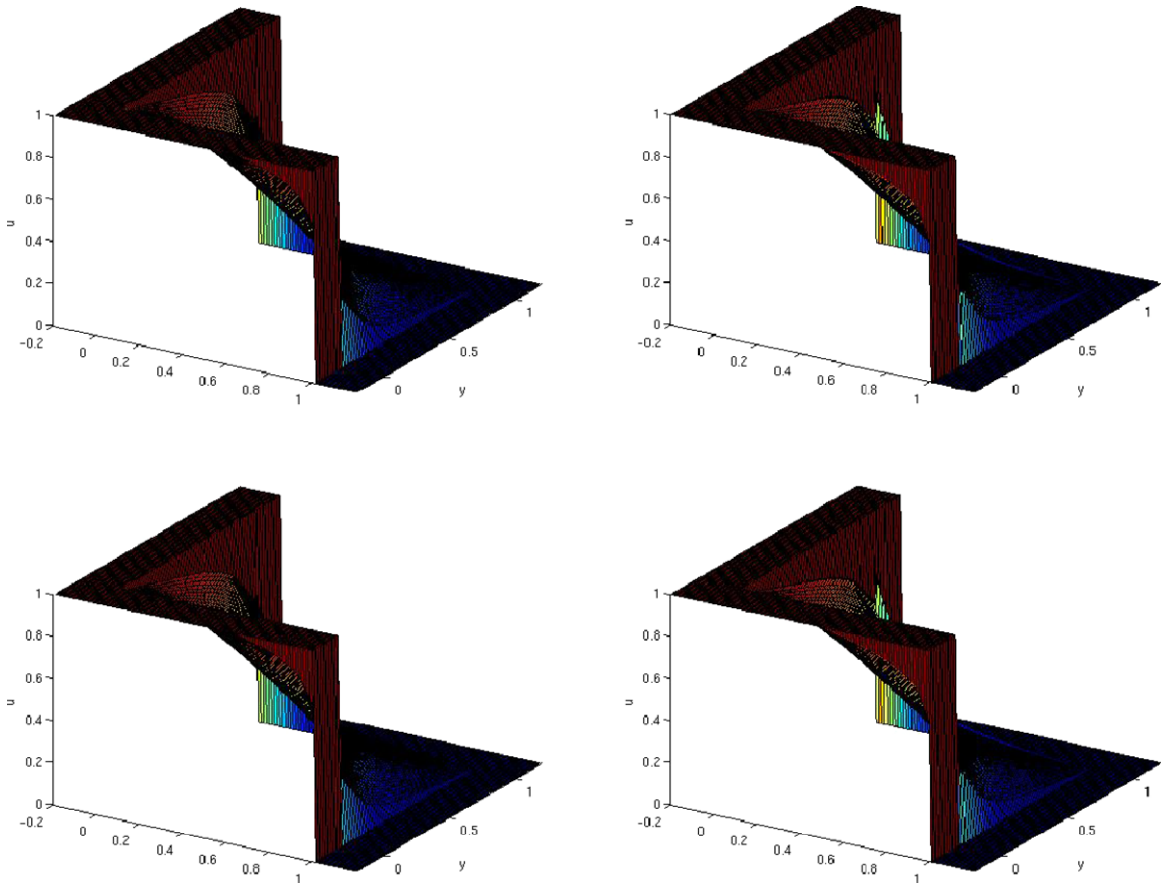


Fig. 5. Finite element solution of the nonlocal models using piecewise constant (left) and piecewise bilinear (right) basis functions on the 80×80 grid in Example 3 with $\epsilon = 1$. Top: the upwind model, bottom: the central model.

the solution obtained by piecewise bilinear basis functions loses the maximum principle. The central model is no longer valid since significant unphysical oscillations are produced and it fails to give good numerical approximate solutions with either constant or bilinear element discretizations.

4.2. Experiments for δ -convergence

To investigate the δ -convergence, we need to keep the ratio $K = \frac{\delta}{h}$ fixed. As the horizon goes to zero, the solutions are expected to converge to that of the corresponding classical local model, if convergent.

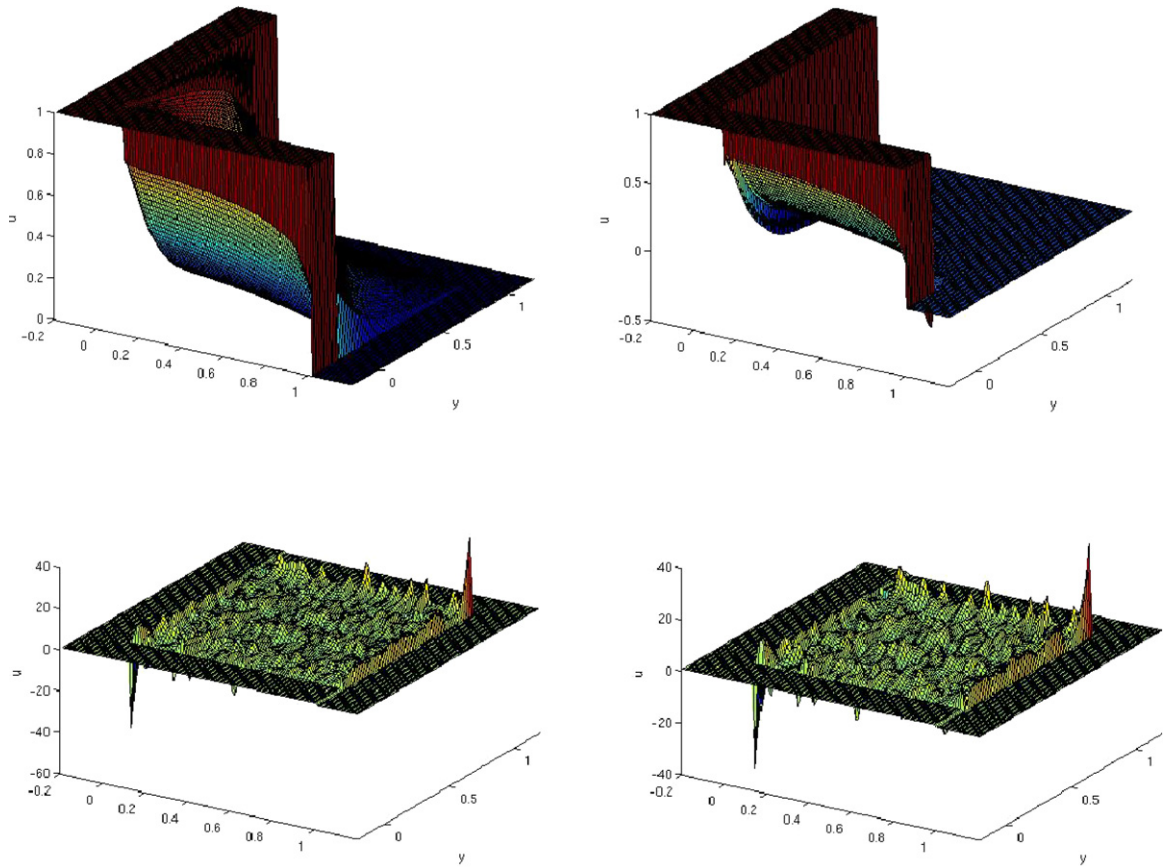


Fig. 6. Finite element solution of the nonlocal model using piecewise constant (left) and piecewise bilinear (right) basis functions in Example 3 on the 80×80 with $\epsilon = 10^{-5}$. Top: the upwind model, bottom: the central model.

Example 4. In this example, we take the exact solution and coefficients of the nonlocal convection–diffusion model as those in Example 1. The right-hand-side function is given by the limit local case. The horizon is no longer fixed, we take $\delta = 20h$.

We solve the nonlocal problem (2.1) over the $N \times N$ square meshes with $N = 40, 80, 160, 320$. It is known [17,42] that the numerical solution of the pure nonlocal diffusion problem obtained by the finite element method with piecewise constant basis functions converges to a wrong limit. We find that the same phenomenon does appear here and the finite element approximate solutions of both nonlocal convection–diffusion models fail to converge to $u(x, y) = x^4 y^2$ when the finite element with piecewise constant basis functions is applied.

In Table 5 numerical results on the L_2 errors and convergence rates of the upwind model by piecewise bilinear basis function discretization are reported. It is easy to see that for the case of $\epsilon = 1.0$, the convergence rates smoothly decrease from second order to first order as the mesh size h gets smaller, while for the case of $\epsilon = 10^{-5}$, the convergence rates are basically of first order. To understand this, we need to refer to (2.6) and (2.7) for the nonlocal upwind model. The nonlocal diffusion term converges to the classic local diffusion counterpart with an order of $O(\delta^2) = O(h^2)$ for $\delta = Kh$, but the nonlocal convection term converges to the local convection with an order of $O(h)$. As for the convection dominated case, the convergence behavior of the convection part dominates the whole process, so the convergence is basically of first order. As for the case of $\epsilon = 1$, we believe that because the diffusion term converges with a higher order rate, the errors of the convection term gradually dominate the convergence process and the rate finally reduces to first order with further mesh refinement.

In Table 6, we present the test results for the central model. Only the case for $\epsilon = 1.0$ is listed since the finite element method with piecewise bilinear basis functions fails to converge for the convection-dominated case. This is consistent with our conclusion on stability. When the method does converge, we find that the convergence rates are

Table 5

Numerical results on errors and convergence rates of finite element approximations of the upwind nonlocal model using piecewise bilinear basis functions in Example 4.

$\delta = 20h$				
h	$\epsilon = 1.0$		$\epsilon = 10^{-5}$	
	p.w. bilinear		p.w. bilinear	
	$\ e^h\ _{L^2}$	CR	$\ e^h\ _{L^2}$	CR
1/40	6.3133e-02	–	2.6396e-01	–
1/80	1.8106e-02	1.81	1.0734e-01	1.2981
1/160	6.5798e-03	1.46	5.2209e-02	1.0398
1/320	2.5761e-03	1.35	2.6807e-02	0.9617

Table 6

Numerical results on errors and convergence rates of finite element approximations of the central nonlocal model using piecewise bilinear basis functions in Example 4.

$\delta = 20h$			
h	$\epsilon = 1.0$		CR
	p.w. bilinear		
	$\ e^h\ _{L^2}$		
1/40	3.3112e-02		—
1/80	6.7850e-03		2.29
1/160	1.3720e-03		2.31
1/320	3.7035e-04		1.89

basically of second order as expected. According to (2.6) and (2.10) for the central model, both the diffusion and convection terms converge to their local counterparts with a rate of $O(h^2)$.

5. Conclusions

Nonlocality is ubiquitous in nature. Nonlocal models are thus of interests. However, historically much more studies have been focused on local PDE models as they do not have to account for the more complicated nonlocal interactions if these are insignificant. With the growing interests in modeling complicated processes in highly heterogeneous environment, nonlocal models can better capture the physical nature than the classical PDE models especially when the latter are not well-posed or their solutions cannot describe the singular behavior in the solutions.

In this paper, we present two models of convection–diffusion processes in the nonlocal setting. They can be seen as generalization of the local convection–diffusion equations. Yet depending on how the nonlocal interactions take place, we see that some (the upwind model) provide more physical descriptions while some other (the central model) may produce unphysical oscillations. The former also allow numerical approximations that are more stable and robust even if we have a convection-dominated problem. The current study is a first step in this direction as we are interested in examining its nonlinear version. In particular, it is of practical value to study the nonlocal nonlinear models in the presence of highly singular solutions. By avoiding spatial differentiations explicitly, nonlocal models have the advantage of creating a more natural setting to study singularities. The current study on the finite element approximations serves as a basis for future development of numerical methods for more complicated nonlocal models, such as models with more generic kernel functions as well as three-dimensional and dynamic models.

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